

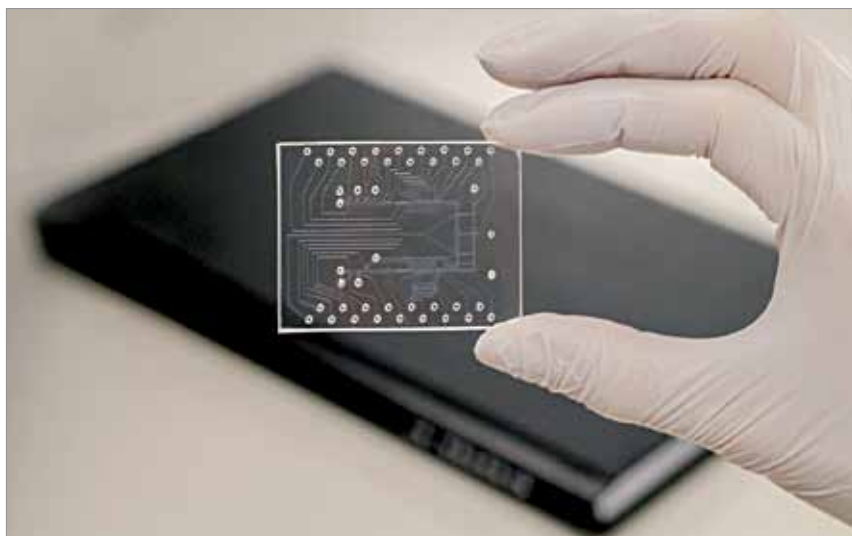


Fig. 3: A microfluidic chip

the patient. These portable devices, equipped with MEMS sensors, can rapidly detect cancer biomarkers, allowing for early diagnosis and immediate treatment decisions in remote or resource-limited settings. The integration of MEMS sensors with microfluidics enables precise and automated sample handling, improving the efficiency and accuracy of point-of-care cancer diagnostics.

4. Tissue analysis. MEMS sensors can be used in tissue analysis devices to assess cellular characteristics and identify cancerous tissues. These sensors enable real-time monitoring of cellular properties such as mechanical stiffness, electrical impedance, or refractive index. By analysing these properties, MEMS-based tissue analysis devices can differentiate between healthy and cancerous tissues, aiding in tumour localisation and surgical guidance.

5. Wearable sensors. MEMS sensors can be incorporated into wearable devices to continuously monitor physiological parameters related to cancer such as heart rate, respiratory rate, or body temperature. These sensors enable long-term monitoring and early detection of cancer associated symptoms or physiological changes. Continuous

monitoring using wearable MEMS sensors provides valuable data for disease management, treatment optimisation, and personalised care.

6. Lab-on-a-chip platforms.

MEMS sensors are integral components of lab-on-a-chip platforms, which integrate multiple diagnostic functions onto a single chip. These platforms allow for the miniaturisation and automation of complex laboratory procedures, enabling rapid and cost-effective cancer diagnostics. MEMS sensors integrated into lab-on-a-chip devices contribute to precise sample handling, accurate detection, and real-time monitoring of cancer-related biomarkers.

MEMS sensors offer significant potential in cancer diagnostics devices. Their high sensitivity, miniaturisation capabilities, and compatibility with microelectronics enable the development of advanced and portable devices for early cancer detection, liquid biopsies, point-of-care testing, tissue analysis, wearable monitoring, and lab-on-a-chip platforms. By harnessing the power of MEMS technology, cancer diagnostics can be enhanced, leading to improved patient outcomes, personalised treatment approaches, and better management of the disease. **EFY**

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Branding is what people say about you when you are not in the room.

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-Jeff Bezos

